Exam

Advanced Macroeconomics

Winter 2023/2024

Problem Set #n

Solutions :

Question 1 .

Downloading the Data Set :

First I navigate to the “Eurostat” database , than I choose “Detailed dataset” > “Economy and finance” > “National accounts (ESA 2010) (na10)” > “Quarterly national accounts (namq_10)” > “Main GDP aggregates (namq_10_ma)” > “GDP and main components (output, expenditure and income) (namq_10_gdp)” .

Next I open the data and start putting filters to my data and selecting the variables , unit , adjustment , periods and the country which in my country is **Greece** .

To the selection part I select only the variables that I need and that are mentioned in the table of the exam pdf. I select the rows and the columns that I want my table to have . As Row I select the time periods and as columns I select the UNIT and Variables . Then I download the data in excel format. After downloading it I do the preparation of the data in excel in order to be clear and to contain what I need .

1.I change the number format with removing the comma (,) from the numbers:

- I select the cells containing the numbers with commas.
- I right-click and choose "Format Cells."
- In the Number tab, I select "Number" from the list.
- I uncheck the "Use 1000 Separator (,)" option. Click "OK."

2.I change the date format from “2005-Q1” to “2005Q1”:

- I select the cells containing the dates.
- I press Ctrl + H to open the "Find and Replace" dialog.
- In the "Find what" box I type - .
- I leave the "Replace with" box empty to remove the dash line.
- Click "Replace All."

Then I write the labels names as : **“Period” “GDPnom_CP” “GOVnom_CP” “GDPnom_PD10” “GOVnom_PD10”.** I delete the extra sheets and columns I don’t need and I save the file as .csv file in order to use it in Octave .

Explain the variables :

B1GQ in the table represents GDP at market prices (nominal).

- **PC_MEUR** (Current prices, million euros): When B1GQ is measured in PC_MEUR, it means that the GDP is reported in current prices unadjusted for inflation or price changes , expressed in million euros. This variable I named it in my file as **"GDPnom_CP".**

- **PD10_EUR** (Price index or deflator in euros): When B1GQ is measured in PD10_EUR it refers to a price index known as the deflator for GDP. This is a measure of overall inflation from the base year of 2010.The deflator is used to convert the current price measures of GDP into constant prices. This variable in named it in my file as **"GDPnom_PD10"**

P3_S13: It represents the final consumption expenditure of government (nominal). It is a measure of the total spending by the government within the period.

- **PC_MEUR** : When P3_S13 is measured in PC_MEUR, it means that the government's consumption expenditure is reported in current prices.This variable I named it in my file as **"GOVnom_CP"**

- **PD10_EUR** (Price index (implicit deflator), 2010=100, euros): It is a specific price index for government spending. It provides a measure of inflation for the basket of goods and services purchased by the government. This variable I named it in my file as **"GOVnom_PD10".**

Question 2

[1]To start with my solution I first use an initial setup with the "disp" function where it allows me to display the "Problem Set n" solutions. This is useful for indicating the start of a problem set in my output.Then I close and I clear the environment which is useful for clearing any previously generated plots , variables and functions.

Then I create flags and I set them to 1 which means "True" . I indicate that data should be loaded into the workspace. I suggest that variables should be defined for use in the problem set.I control whether a plot should be generated , then determining that these plots should be saved as files (pdf) . Last I Indicate that correlation calculation should be performed.With these steps I set up a controlled environment.

I check for the matlab or octave environment (in my case octave) to adjust the execution of code based on the detected environment. After this I Initialize a figure counter which is useful to generate multiple figures. I define a color palette for plots and I set the line width for clear visual presentation of plotted data which makes sure that the lines in plots are enough thick.

[1] Advanced Macroeconomics Exercises : Tutorial 5 (Plot_FRED_Data.m)

[2]After checking the script is running in Octave the code loads two packages needed for octave : io and dataframe. These packages are necessary for input and output operations and handling data in dataframe format. The " **if LoadData**" condition checks if the script should continue with loading data , which is set earlier . I use the dataframe function to load the csv file named eurostatFile.

To initiale the time variable I first set the first year to 2005 which is the first year for my data . Then I calculate the number of observations in the dataset based on the length of one column of my EUROSTAT data . The **Time** function creates a time vector starting from the first year of my data , including all the quarters observations and ending up to the final year. The **transpose** function makes sure to convert the row vector which is needed for time series data .

[3]I prepare my dataset for further analysis by defining variables that will be used . I extract the specific columns from the EUROSTAT dataframe and assign them to descriptive named variables : GDPnom , GOVnom , GDPDeflator , GOVDeflator.

Question 2 (a)

[4]Calculating the real values of GDPnom and GOVnom:

The purpose here is to calculate the real values adjusted for inflation.

Real GDP represents the total value of all goods and services produced by an economy in a given period adjusted for inflation which reflects the economy's true growth without the effect of price level changes.

Real Government Spending measures the government's expenditure on goods and services, adjusted for inflation which provides insight into the government's contribution to the economy independent of changing price levels.

First I create variables named "**GDP_real"** and **"GOV_real"** to store the calculated real values for GDP and government spending.I equal these with dividing the nominal values by their corresponding deflators and then multiplying with 100. Here the deflators adjust the nominal values to remove the effects of price changes over time.

Question 2 (b)

[5]Applying the hp filter :

This is about transforming real GDP and real government spending data to analyze their cyclical components using the Hodrick-Prescott (HP) filter. The first step involves calculating the natural logarithm (log) of the real GDP (GDP_real) and real government spending (GOV_real).

Then I apply the Hodrick-Prescott (HP) filter to the logged real GDP and government spending data , which is used to separate time series into trend component and cyclical component.

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[2] Advanced Macroeconomics Lecture : Lecture 7 (octxmpl_7_m).
[3] Advanced Macroeconomics Exercises : Tutorial 5 (Plot_FRED_Data.m)
[4] Advanced Macroeconomics Lecture : Lecture 7 (octxmpl_7_m).

[5] Advanced Macroeconomics Lecture : Lecture 7 (octxmpl_7_m).

[6]I choose the smoothing parameter (lambda) equal to **1600**, which is a standard value for quarterly data . (The larger the parameter the smoother the trend) .

Question 2 (c)

[7]Here I use the **corr** function is used to compute the correlation coefficient between the cyclical components of real GDP (GDP_cycle_real) and real government spending (GOV_cycle_real). A coefficient close to 1 means a strong positive correlation, meaning that when one variable increases, the other does so too . When displaying the coefficient **" disp(Coef_CORR)"** it is equal to = 0.5682.

Question 2 (d)

[8]First I calculate the ratio of nominal government spending **(GovNom)** to nominal GDP (**GDPnom)** for each time period in the dataset, multiplying this ratio by 100 converting it into a percentage. After GOV/GDP ratio is calculated for each time period, than I calculate the average value using the mean function.

The average ratio gives a single value that is the typical proportion of government spending from GDP across the entire dataset.Then I display the computed average ratio in the Octave command window.

Question 2 (e)

[9]I start by Finding the Index for the Year 2010 where I identify the indices within the **Time** vector that correspond to the year 2010. The condition that I add **Time >= 2010 & Time < 2011** ensures that all quarters within 2010 are included.

Next I calculate the average real GDP "**(baseGDP_2010)"** and real government spending "**(baseGOV_2010)"** for the year 2010, using the index identified in the previous step. If there are multiple entries for 2010 this calculates their mean.

Finally I convert the entire real GDP and real government spending series to indices based on the 2010 base values. I divide each real value in the series by the respective 2010 base value and then multiply by 100. This process creates a new series where the value for 2010 is 100.

Question 3

i. Plot the indexed time series of real GDP and government spending for all quarterly observations.

I start plotting with the condition that "**(PlotOverview)"** is met allowing to control when visualizations are generated. I insert a figure counter and create a new figure window for the plots making sure that each plot appears in a separate window .

I insert in the code "**subplot"** which divides the figure into two sections (2 rows) .The first subplot displays the indexed real GDP over time and the second subplot shows the indexed real government spending. For each subplot I use a different color for differentiation and is labeled with appropriate axis titles and labels for each of them .

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[6] Advanced Macroeconomics Lecture : Lecture 7 (Chapter – 7 .pdf , page 7)
[7] Advanced Macroeconomics Exercises : Tutorial 5 (Plot_FRED_Data.m)
[8] Advanced Macroeconomics Lecture : Lecture 7 (octxmpl_7_m)
[9] Advanced Macroeconomics Lecture : Lecture 7 (octxmpl_7_m)

Then I save both of the subplots into one pdf with the respective name as in the code file .

For :

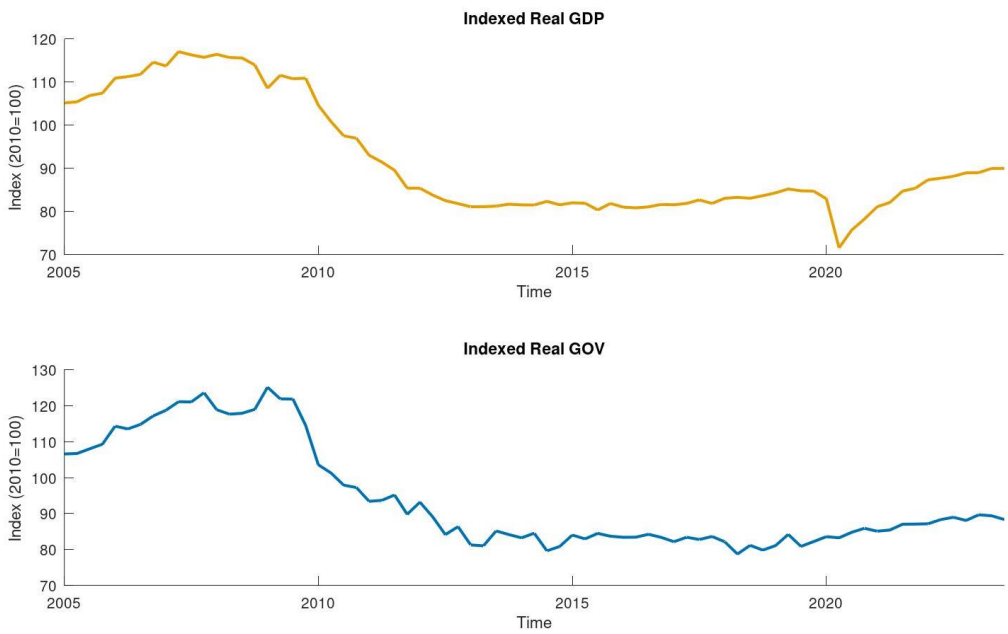
i. The plot of public consumption to GDP ratio and its average in percent.
ii. The plot of the cyclical components.

[10]I repeat the same procedure as above for the first plot but with choosing the respective variable I want to plot , changing the x , y labels and changing the title of the plot .

For plot (iii) I also add a scatter plot of the cyclical components of GDPreal and GOVreal where I use the command "**scatter"** which creates the plot with the cyclical components of real GDP (GDP_cycle_real) on the x-axis and the cyclical components of government spending (GOV_cycle_real) on the y-axis. The **'filled'** option specifies that the dots in the scatter plot should be filled , which makes it easier to read .

Question 4 : Given these plots, describe the development and relationship of the presented variables referring to special events in the past.

Plot (i)



The first plot shows the indexed real GDP and government spending for Greece with both series rebased to 100 in the year 2010. This graphically represents how the real GDP and government spending have changed over the years relative to their respective values in 2010.

From the plot, we can observe the following:

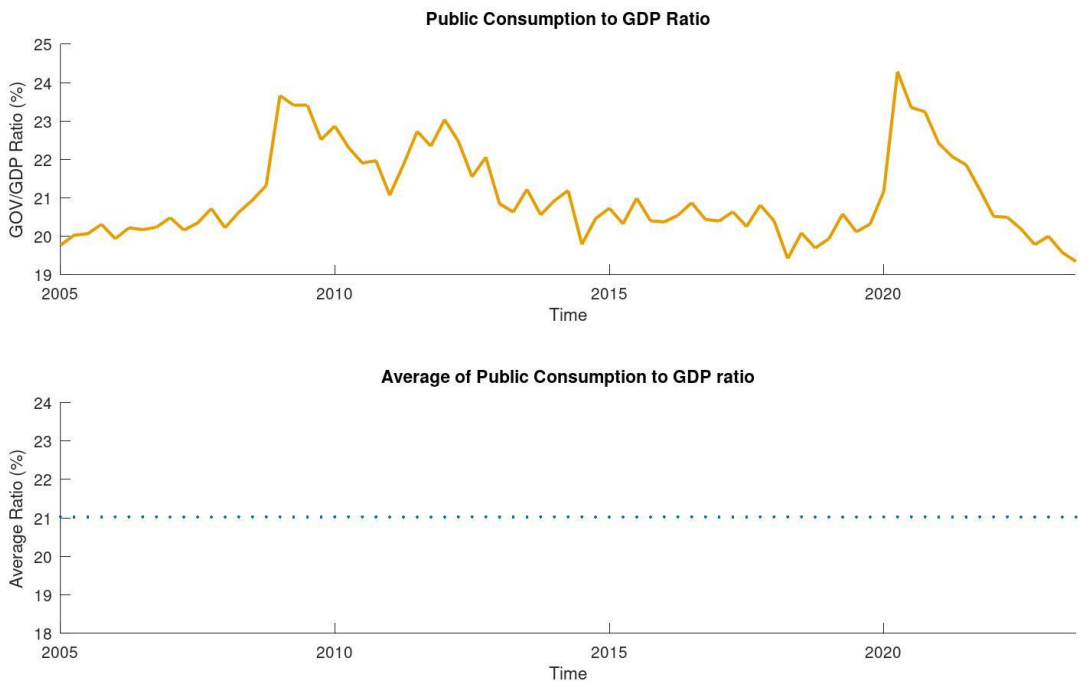
[10] Advanced Macroeconomics Exercises : Tutorial 5 (Plot_FRED_Data.m)

- [11]The indexed real GDP appears to have significantly decreased after 2010, which could be related to the Greek government-debt crisis and resulting economic reforms and austerity measures. (Wikipedia , 2023).

- [12]The indexed government spending shows a a significant drop after 2010, which was probably caused by the spending reductions that were part of the austerity measures.The austerity measures introduced in Greece were a response to a big borrowing done after the international financial crisis.These actions included raising taxes, cutting back on spending, and reducing pensions and benefits.(Britannica, 2023).

- [13]Both series show that there has been modest improvement in both series recently, suggesting a partial economic recovery. Greek economy showed modest progress after the end of the debt crisis as a result of initiatives to promote growth and reduce austerity restrictions. (Wikipedia , 2023).

Plot (ii)



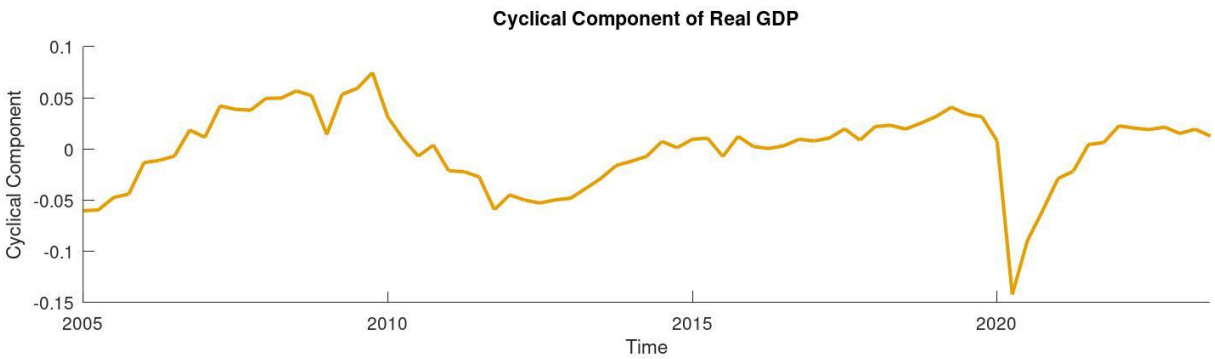
[11] (https://en.wikipedia.org/wiki/Greek_government-debt_crisis)

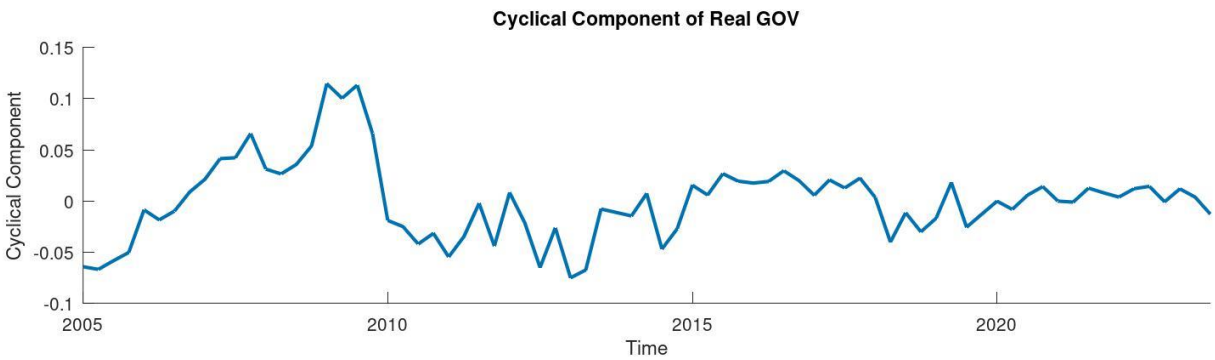
[12] (https://www.britannica.com/place/Greece/Greeces-debt-crisis)

[13] (https://en.wikipedia.org/wiki/Greek_government-debt_crisis)

The plot shows the public consumption to GDP ratio for Greece and its average , which show better the austerity measures that were introduced after 2010.
[14]Greece hosted the Summer Olympics in 2004, which resulted in large public spending and an increase in the deficit for the nation. The ratio reached its peak around 2011, which is the period of intense economic austerity measures in Greece. (Council on Foreign Relations.)
This peak corresponds to a period of relatively high public consumption, probably as a result of government initiatives to boost the economy. From the graph we see some recovery for the years 2015 and on where the ratio declined as the Greek economy began to stabilize and recover.[15]But it then started to rise once more in 2020 as a result of more government spending in reaction to the crisis of Covid-19 Pandemic . (CEPR,2021)

Plot (iii)





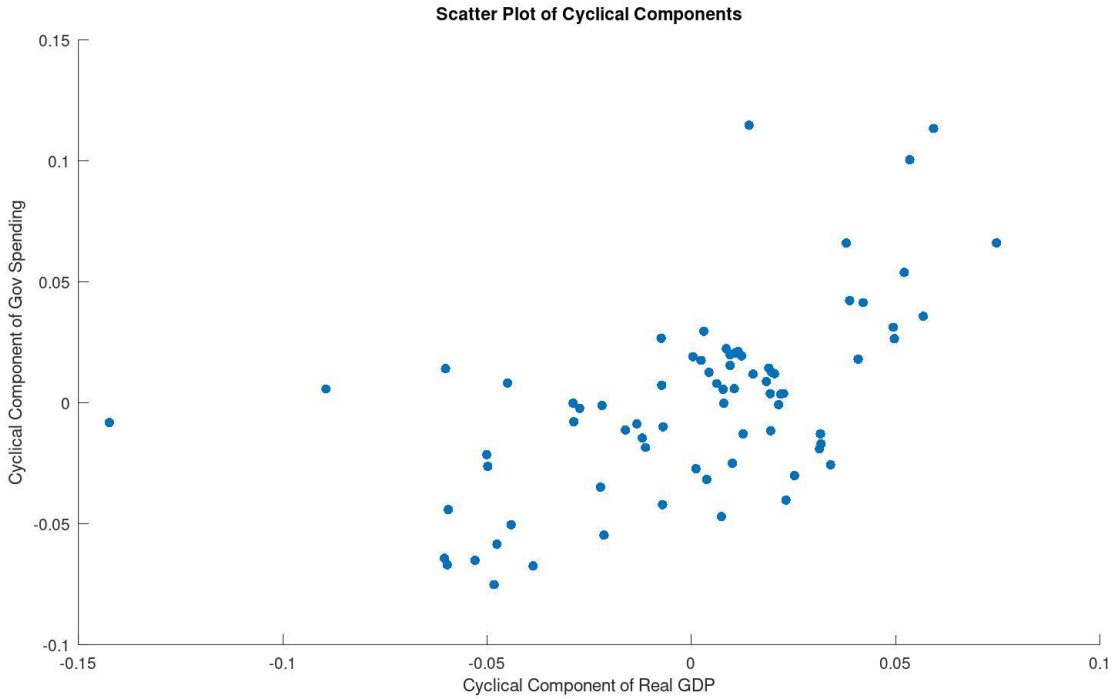
The cyclical components plot for real GDP and government spending in Greece offers insights into the economic fluctuations the country experienced, particularly during and after the global financial crisis .

[14] https://www.cfr.org/timeline/greeces-debt-crisis-timeline

[15] https://cepr.org/voxeu/columns/pandemic-and-greeces-debt-day-after

Real GDP Cyclical Movements: The cyclical movements of Greece's real GDP between 2007 and 2017 reflect a notable economic contraction. This decline was a result of the Greek depression which was intensified by policy measures such as tax increases. **Cyclical Component of Real Government Spending**:, The graph's variations for actual government spending are consistent with the bailout (recovery) period of austerity measures. As part of the fiscal consolidation measures, tax increases and spending reductions were implemented to reduce Greece's government expenditure.[16](Becker Friedman Institute , 2021)

Plot (iv)



Interpretation :

The scatter plot shows the relationship between the cyclical components of government spending and Real GDP in Greece.

Points that are grouped around the origin suggest periods where both the Real GDP and government spending were close to their long-term trends. As the points spread out, they show the deviations from these trends. Since the points do not create a distinct line pattern, the distribution of points does not show a strong linear link between the two variables.

The fact that there are times when both indicators are above or below their trends and times when one is above and the other is below points to a complicated relationship between the

[16] https://bfi.uchicago.edu/insight/research-summary/the-macroeconomics-of-the-greek-depression/

trends in government spending and GDP. Points are spreaded which indicates that other factors also influence the cyclical components of GDP and government spending.

[17]**Question 5** :

Solution :

Given Non-linear Constraint:

- $Y_t = C_t + G_t$

Given Relationship for Public Consumption:

- $Gt = \tau_t Y_t$

- $g_t := ln(G_t)$

Goal:

To show that the log-linearized constraint can be written as $\mathbf{y_t = c_t + g_t}$ using the Taylor series approximation for log-linearization.

Steps :

1. **Logarithm of the Non-linear Constraint:**

Since we cannot take the logarithm of a sum directly we express $\mathbf{G_t}$ in terms of $\mathbf{Y_t}$ using the given relationship, and then we take the log of both sides :

$\ln(Y_t) = \ln(C_t + G_t)$

$\ln(Y_t) = \ln(C_t + \tau_t Y_t)$

2. **Expressing Gt in Log Terms:**

Given the relationship $g_t \equiv -\ln(1-\tau_t)$, we have τ_t as a function of g_t :

First I exponentiate both sides : $e_{gt} = 1/_{1-\tau_t}$

And then we can write it : $\tau_t = 1 - e^{-gt}$

And so $\mathbf{G_t}$ in terms of $\mathbf{Y_t}$ is: $\mathbf{Gt = (1 - e^{-gt}) Yt}$

[17] Zietz, Joachim (2006): Log-Linearizing around the Steady State: A Guide with Examples. Middle Tennessee State University. (https://papers.ssrn.com/sol3/papers.cfm?abstract_id=951753)

[18]Taylor Series Approximation:

- I use the Taylor series approximation for a function $g(X,Y)$ around the steady state.
- For a function $f(x)=ln(1-x)$ the first-order Taylor series expansion around a point $x=0$ is: $f(x)\approx f(0)+f'(0)\cdot x$
- Since $f(x)=ln(1-x)$, we have $f'(x)=-1/1-x$ and at $x=0$, $f'(0)=-1$. So $\ln(1-x)\approx -x$
- Applying this to g_t expression we approximate τ_t as g_t: $\tau_t=1-e^{-gt}\approx g_t$
- So τ_t which is $1-e^{-gt}$, is approximately gt when gt is small. $\tau_t\approx g_t$

Therefore G_t can be written as : $\boldsymbol{G_t= g_t Y_t}$

[19]Constructing the Log-Linearized Equation:

- I substitute G_t with $g_t Y_t$ in the logarithmic form:

$$\ln(Y_t)=\ln(C_t+G_t)$$

$$\ln(Y_t)=\ln(C_t+ g_t Y_t)$$

- Since we're assuming that g_t represents a small deviation, we can add the logs :

$$\ln(Y_t)\approx\ln(C_t)+\ln(g_t Y_t)$$

- Because g_t is already the log of G_t representing the proportion of output that is government spending, we simplify further:

$$\ln(Y_t)\approx\ln(C_t)+g_t$$

- After subtracting $\ln(Y_t)$, we get: $0\approx(\ln(C_t)-\ln(Y_t))+g_t$

- The goal of log-linearization is to express variables as deviations from their steady states. If we denote the steady state of Y_t as $\bar{Y}$, the deviation of Y_t from its steady state in logarithmic terms is $y_t=\ln(Y_t)-\ln(\bar{Y})$.

- We apply the same for Ct until we are left with the equation : **yt=ct+gt**

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[18] Advanced Macroeconomics Lecture : Chapter 4 , Page 6 (Taylor Approximation)
Advanced Macroeconomics Lecture : Chapter 7 , Page 36 (Taylor Approximation)

[19] Advanced Macroeconomics Lecture : Chapter 7 , page 37 (Log-Linearization)
Advanced Macroeconomics Tutorial : Exercises Solution 7 .

Question 6 :

- **Expression for the natural interest rate**

I start by by defining an expression for the natural interest rate r^n_t.

[20]To do this I start by rearranging the linearised consumption Euler equation with regard to real interest rate rt which is equal to $= i_t - E_t\pi_{t+1}$.

$c_t = E_t c_{t+1} - 1/\theta\,(i_t - E_t\pi_{t+1} - \rho)$ **Equation (i)**

$* \; -1/\theta\,[i_t - E_t\pi_{t+1} - \rho] = c_t - E_t c_{t+1}$

$* \; -[i_t - E_t\pi_{t+1} - \rho] = \theta\,[\,c_t - E_t c_{t+1}\,]$

$* \; \underbrace{i_t - E_t\pi_{t+1}} - \rho = -\,\theta\,[\,c_t - E_t c_{t+1}\,]$

Since $r_t = i_t - E_t\pi_{t+1}$, then we have :

$r_t = \rho + \theta[(E_t\, c_{t+1}) - c_t\,]$ **Equation (ii)**

Since our model is :

$yt = ct + gt$

,the change in output from period t to period t+1 would be equal to the change in consumption plus the change in government spending over the same period. Therefore the equation can be written in difference form as : $yt+1-yt=(ct+1-ct)+(gt+1-gt)$

Here we can express $(ct+1-ct)$ from this equation and imply it to the real interest rate equation **(ii)** $r_t = \rho + \theta[(E_t\,\mathrm{yt+1}) - \mathrm{yt} - (E_t\mathrm{gt+1}) + \mathrm{gt}\,]$

Applying the natural real interest rate on natural output and government spending we have :

$r_t^n = \rho + \theta\,[(E_t y^n{}_{t+1}) - y^n{}_t - (E_t g_{t+1}) + g_t)]$ **Equation (iii)**

- **Imputing the natural real interest rate into linearised consumption Euler equation**

-From Equation **(iii)** I take out ρ as $\rho = r_t^n - \theta\,[(E_t y^n{}_{t+1}) - y^n{}_t - (E_t g_{t+1}) + g_t)]$.

-Then I input it to the linearised consumption Euler equation **(i)** where we then have :

$c_t = E_t c_{t+1} - 1/\theta\,(i_t - E_t\pi_{t+1}) + 1/\theta\,\{r_t^n - \theta\,[(E_t y^n{}_{t+1}) - y^n{}_t - (E_t g_{t+1}) + g_t)]\}$

[20] Alogoskoufis, George (2019): Dynamic Macroeconomics, MIT Press,Chapter 14 ,Page 616 . "Fisher Equation"

-Rearranging the largest bracket { } , where for the second bracket [ ] we eliminate θ ,we have:

$$c_t = E_t c_{t+1} - 1/\theta\,(i_t - E_t\pi_{t+1}) + 1/\theta * r_t^n - E_t y^n_{t+1} + y^n_t + E_t g_{t+1} - g_t$$

-Putting into one bracket the terms multiplied by $1/\theta$ we have:

$$c_t = E_t c_{t+1} - 1/\theta\,(i_t - E_t\pi_{t+1} - r_t^n) - E_t y^n_{t+1} + y^n_t + E_t g_{t+1} - g_t$$

-Since we have $y_t = c_t + g_t$ we can express $c_t = y_t - g_t$ and $E_t c_{t+1} = E_t y_{t+1} - E_t g_{t+1}$ and substitute them in the equation :

$$y_t - g_t = E_t y_{t+1} - E_t g_{t+1} - E_t y^n_{t+1} + y^n_t + E_t g_{t+1} - g_t - 1/\theta\,(i_t - E_t\pi_{t+1} - r_t^n)$$

-We can eliminate g_t and $E_t g_{t+1}$ since they are subtracted with one another and put $y^n t$ on the left side :

$$y_t - y^n_t = E_t y_{t+1} - E_t y^n_{t+1} - 1/\theta\,(i_t - E_t\pi_{t+1} - r_t^n)$$

-Since the out put gap is defined as the difference between actual and natural output

$\tilde{y}_t = yt - y^n_t$. Then also $E_t\tilde{y}_{t+1} = E_t y_{t+1} - E_t y^n_{t+1}$.

-We substitute this in the equation above and we get :

$$\tilde{y}_t = E_t\tilde{y}_{t+1} - 1/\theta\,(it - E_t\pi_{t+1} - r_t^n)$$

Question 7 :

[21]To run a stochastic simulation of the model with a shock I start with these steps :

1. I define the model's variables and shocks I want to run .
    - I put the endogenous variables of the model specified from the equations in the Problem Set , which include outputs like actual output, consumption, labor, etc
    - I define exogenous shocks that will be applied to the model, specifically shocks to government spending and a monetary policy shock.
2. I set up the parameters of the model which are all specified in the dynare script of what is their role in the model and the values specified for each.
3. Specifying the model's equations

[21] Advanced Macroeconomics Lecture : Chapter 8 , nkmodel.mod.

- I insert all the equations which describe the economic relationships and policies in the model. These equations are important for simulating the economy's response to various shocks.

4. I set up the initial conditions

   - I specify initial values for each of the variables in my model. These values serve as the starting point for the simulation representing the state of the economy before shocks are applied.

5. Steady state
   - I instruct Dynare to calculate the steady state of the model, which is the condition where the economy is at equilibrium. Here all variables are constant over time since there is no shock.

6. Configure shock stimulation

   - I define the characteristics of the exogenous shock to government spending as specified in the model with a standard deviation stderr; = 0.1. I do the same with the eps_nu , monetary policy shock where I set the stderr=0.003.[22]

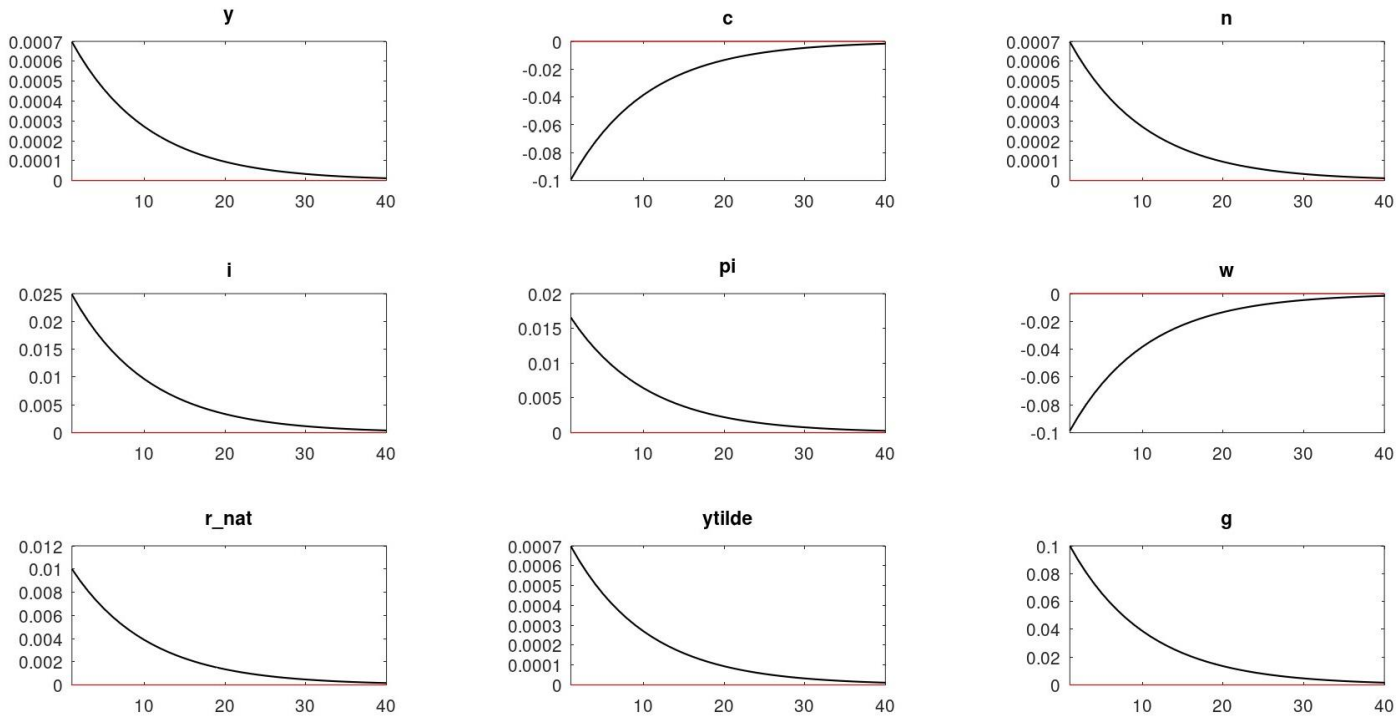
7. I run the Stochastic Simulations

   - These simulate the behavior of the economy over a set number of periods (in my case 200) , with the defined shock.I make sure that the first 100 periods of simulation data will be discarded until to reach the steady state ,and I set a first order approximation of the model. The simulation generates impulse response functions (IRFs), which show the effect of a one-time shock to the variables.

[23]Question 8

The impulse response functions:

[22] Advanced Macroeconomics Lecture : Chapter 8 , nkmodel.mod.
[23] Advanced Macroeconomics Tutorial : Tutorial 9 , AM-RBC-Guide.pdf, pages 8.



The y axis present the deviation from the baseline (steady state) in response to the shock.

Actual Output (y): Equation (1) from the model indicates that output is the sum of consumption (c) and government spending (g). The IRF shows an immediate increase in output from the government spending shock, which means there has been direct impact of increased government spending on output.

Consumption (c): The IRF for consumption shows a decrease. One potential reason could be that when there is an increase in government spending consumers need to pay higher future taxes and so they reduce their consumption .

Labor (n): Equation (2) shows labor (n) is equal to output (y), implying labor should increase with output.The IRF shows an immediate increase just the same as output .

Nominal Interest Rate (i): The IRF shows that the nominal interest rate increases, which could be due to the central bank's response to inflation as indicated in Equation (12).

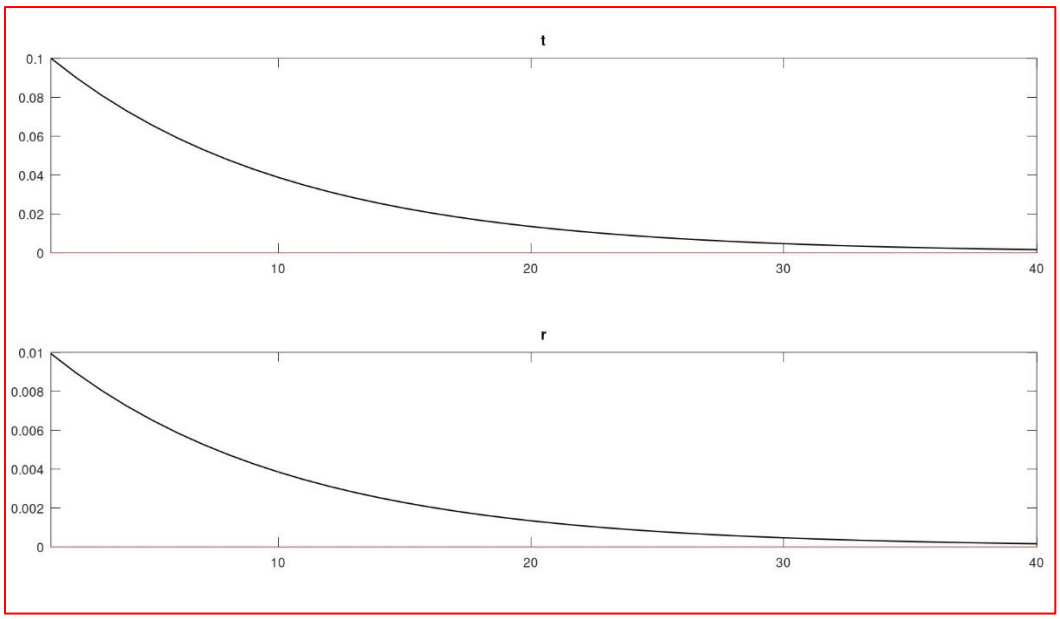
Inflation (pi) : Inflation rises immediately following the shock.This suggests that the increase in government spending creates increase pressure on prices. The New Keynesian Phillips Curve from equation (6) links inflation to the output gap so an increase in the output gap can lead to higher inflation.

Real wage (w) appears to decrease in response to the government spending shock.This might suggest that the increase in labor puts downward pressure on wages, as indicated by equation (5).

Natural Rate of Interest (r_nat):The natural rate of interest increases slightly after the shock.An increase in government spending can lead to an increase in the natural level of output and raise the natural rate of interest , as indicated by equation (4)

Output Gap (ytilde): The IRF for the output gap shows an initial increase. This comes because the direct effect of increased government spending raises actual output above its natural level, as indicated from Equation (9).

Government Spending (g**)**: The IRF shows that government spending increases, as expected, in response to the positive shock eps_g.



Taxes (t): As initial response they increase in response to the government spending shock. This is consistent with equation (6) where taxes are directly linked to government spending, probably because they are used to finance the increase in spending .

Real Interest Rate (r): The real interest rate has a slight increase from the shock. This could be the result of the central bank raising nominal interest rates .

The IRFs presented show that the effect of shock diminishes over time meaning that the economy is adjusting back to its steady state.

Question 9 :

The empirically observed correlation coefficient of government spending and output for Greece : 0.5682.

The coefficient after a stochastic simulation : 0.9334

CORRELATION OF SIMULATED VARIABLES

VARIABLE	y	c	n	i	pi	w	r_nat	ytilde	g
y	1.0000	-0.9324	1.0000	0.9408	0.9591	-0.9313	0.9334	1.0000	0.9334
c	-0.9324	1.0000	-0.9324	-0.9997	-0.9966	1.0000	-1.0000	-0.9324	-1.0000
n	1.0000	-0.9324	1.0000	0.9408	0.9591	-0.9313	0.9334	1.0000	0.9334
i	0.9408	-0.9997	0.9408	1.0000	0.9983	-0.9996	0.9998	0.9408	0.9998
pi	0.9591	-0.9966	0.9591	0.9983	1.0000	-0.9963	0.9968	0.9591	0.9968
w	-0.9313	1.0000	-0.9313	-0.9996	-0.9963	1.0000	-1.0000	-0.9313	-1.0000
r_nat	0.9334	-1.0000	0.9334	0.9998	0.9968	-1.0000	1.0000	0.9334	1.0000
ytilde	1.0000	-0.9324	1.0000	0.9408	0.9591	-0.9313	0.9334	1.0000	0.9334
g	0.9334	-1.0000	0.9334	0.9998	0.9968	-1.0000	1.0000	0.9334	1.0000
t	0.9334	-1.0000	0.9334	0.9998	0.9968	-1.0000	1.0000	0.9334	1.0000
r	0.9282	-0.9999	0.9282	0.9994	0.9955	-1.0000	0.9999	0.9282	0.9999
nu	-0.4832	0.1339	-0.4832	-0.1578	-0.2156	0.1311	-0.1368	-0.4832	-0.1368

Interpretation :

The simulated coefficient (0.9334) differs significantly from the actually observed one (0.5682). The simulated coefficient implies a higher link between government spending and output than what the empirical data for Greece show.This high coefficient connection indicates a bigger impact of government spending on output than is found in reality.

The difference in correlation coefficients indicates that the model may not adequately capture the complex nature of Greece's economy as shown in question 4 when comparing the plots.

It is possible that additional elements in the empirical data are at work to limit the impact of government expenditure on output, such as fiscal multipliers, economic policies, external economic conditions etc that are not fully accounted for in the model.

The simulation takes into account both government spending shocks (eps_g) and and monetary policy shocks (eps_nu)

Government Spending Shocks (eps_g): For example when government spending rises it raises demand for products and services, resulting in higher output. This direct influence would result in a positive correlation between g and y.

Monetary Policy Shocks (eps_nu**)**: Based on the equations specified in the model where the variable nu is implemented in the interest rate value we can say that the monetary policy shock could affect the interest rate (i), which in turn influences investment and consumption, further impacting output.

Empirically Observed Coefficient :

Empirically the moderate positive connection(0.5682) indicates that even though government expenditure has an effect on output, it is not as dominant as the model claims. This could be caused by a variety of factors:

[24]Fiscal Multipliers: The real fiscal multipliers in the Greek economy may be lower than those predicted by the model. In practice, increases in government expenditure may not result in proportional increases in output because of different factors such as tax rates, the state of public finances, debt levels, ect. (Alogoskoufis , 2019).

[24] Alogoskoufis, George (2019): Dynamic Macroeconomics, MIT Press,Chapter 21 , Pages 947-948

Octave (Version : GNU Octave , 8.4.0) Table :

Programs	Description	Source
disp	Displays text or variables to the command window.	https://octave.sourceforge.io/octave/function/disp.html
close	Close figure window(s)	https://octave.sourceforge.io/octave/function/close.html
clear	Clears variables from the current workspace	https://octave.sourceforge.io/octave/function/clear.html
find	Return a vector of indices of nonzero elements of a matrix, as a row if *x* is a row vector or as a column otherwise.	https://octave.sourceforge.io/octave/function/find.html
mean (x)	Compute the mean of the elements of the vector x.	https://octave.sourceforge.io/octave/function/mean.html
log(x)	Compute the natural logarithm, $\ln(x)$, for each element of x.	https://octave.sourceforge.io/octave/function/log.html
hold on	Retain plot data and settings so that subsequent plot commands are displayed on a single graph. Line color and line style are advanced for each new plot added.	https://docs.octave.org/v8.4.0/Manipulation-of-Plot-Windows.html
hold off	Restore default graphics settings which clear the graph and reset axes properties before each new plot command.	https://docs.octave.org/v8.4.0/Manipulation-of-Plot-Windows.html
subplot (rows, cols, index)	Set up a plot grid with rows by cols subwindows and set the current axes for plotting to the location given by index.	https://octave.sourceforge.io/octave/function/subplot.html
ylabel, xlabel, title	Set the y-axis label, x-axis label, and title of the current axes, respectively.	https://docs.octave.org/v8.4.0/Plot-Annotations.html#XREFylabel
xlim	Query or set the limits of the x-axis for the current plot.	https://octave.sourceforge.io/octave/function/xlim.html
scatter	Draw a 2-D scatter plot.	https://octave.sourceforge.io/octave/function/scatter.html
set	Set named property values for the graphics handle	https://octave.sourceforge.io/octave/function/set.html

pkg load	The functions from a package can be added to the Octave path	https://docs.octave.org/v8.4.0/Using-Packages.html
length (a)	Return the length of the object a.	https://octave.sourceforge.io/octave/function/length.html
transpose (x)	Return the transpose of x.	https://octave.sourceforge.io/octave/function/transpose.html
corr (x)	Compute matrix of correlation coefficients.	https://octave.sourceforge.io/octave/function/corr.html
print	Print a plot, or save it to a file.	https://octave.sourceforge.io/octave/function/print.html
addpath	Add named directories to the function search path	https://octave.sourceforge.io/octave/function/addpath.html
dynare	Runs the dynare file	https://www.dynare.org/resources/quick_start/#running-and-editing-a-dynare-example-on-windows
hpfilter	Applies the Hodrick-Prescott filter to decompose a time series into its trend and cyclical components.	Advanced Macroeconomics Lecture : Chapter 7 , (hpfilter.m)
orient	Query or set the print orientation for figure hfig	https://docs.octave.org/latest/Printing-and-Saving-Plots.html
hf	Manipulate the figure's properties and save it to a file	https://docs.octave.org/latest/Multiple-Plot-Windows.html
*	Multiplication	https://docs.octave.org/v6.2.0/Arithmetic-Ops.html
./	Performs element-wise division on arrays	https://docs.octave.org/v6.2.0/Arithmetic-Ops.html

Dynare (6.0) Table :

var;	Declares endogenous variables in the model	https://www.dynare.org/manual/the-model-file.html Section4.2
varexo;	Declares the exogenous variables in the model	https://www.dynare.org/manual/the-model-file.html Section 4.2
parameters;	Declares parameters used in the model	https://www.dynare.org/manual/the-model-file.html Section4.2
model;	The model is declared inside a model block	https://www.dynare.org/manual/the-model-file.html Section4.2.1
initval;	Provides initial guess values for the non-linear solver in the steady state computation	https://www.dynare.org/manual/the-model-file.html Section 4.7
shocks	Specifies temporary changes in the value of exogenous variables	https://www.dynare.org/manual/the-model-file.html Section 4.8
steady;	Computes the steady state of a model using a nonlinear Newton-type solver	https://www.dynare.org/manual/the-model-file.html Section 4.10.1
stoch_simul;	Computes a Taylor approximation of the model around the deterministic steady state and solves of the the decision and transition functions for the approximated model.	https://www.dynare.org/manual/the-model-file.html Section 4.13.1

Data Section :

[25]My data for country Greece is taken from Eurostat Database which provides quarterly data based on the European System of Accounts (ESA 2010) framework. It includes a variety of economic variables such as government spending, investments, and consumption.

The data I use has the SCA adjustement which in order to provide appropriate comparisons across time periods it is adjusted for seasonal and calendar influences. Since seasonal fluctuations can impact GDP and government spending across years it is very important to provide appropriate comparisons across time periods.

The time frame for my data is in periods from 2005-Q1 to the Latest available which in my situation is the third quarter of 2023.

namq_10_gdp: This is the code for the dataset I use. It stands for National Accounts Main aggregates for Quarterly data, focused on Gross Domestic Product (GDP) and related components.The variables that I use are explained in more detail in the solution of Question 1.

[25]https://ec.europa.eu/eurostat/databrowser/view/namq_10_gdp/default/table?lang=en&category=na10.namq_10.namq_10_ma

References :

https://ec.europa.eu/eurostat/databrowser/view/namq_10_gdp/default/table?lang=en&category=na10.namq_10.namq_10_ma

https://en.wikipedia.org/wiki/Greek_government-debt_crisis

https://www.britannica.com/place/Greece/Greeces-debt-crisis

https://www.cfr.org/timeline/greeces-debt-crisis-timeline

https://cepr.org/voxeu/columns/pandemic-and-greeces-debt-day-after

https://bfi.uchicago.edu/insight/research-summary/the-macroeconomics-of-the-greek-depression/

Zietz, Joachim (2006): Log-Linearizing around the Steady State: A Guide with Examples. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=951753

Alogoskoufis, George (2019): Dynamic Macroeconomics, MIT Press,Chapter 14 ,Page 616 . "Fisher Equation"

Alogoskoufis, George (2019): Dynamic Macroeconomics, MIT Press,Chapter 21 , Pages 947-948

Advanced Macroeconomics Lecture : Chapter 8 , nkmodel.mod.

Advanced Macroeconomics Lecture : Lecture 7 (octxmpl_7_m).

Advanced Macroeconomics Exercises : Tutorial 5 (Plot_FRED_Data.m)

Advanced Macroeconomics Lecture : Lecture 7 (Chapter 7 .pdf , page 7)

Advanced Macroeconomics Lecture : Chapter 4 , Page 6 (Taylor Approximation)

Advanced Macroeconomics Lecture : Chapter 7 , Page 36 (Taylor Approximation)

Advanced Macroeconomics Lecture : Chapter 7 , page 37 (Log-Linearization)

Advanced Macroeconomics Tutorial : Exercises Solution 7 .

Advanced Macroeconomics Tutorial : Tutorial 9 , AM-RBC-Guide.pdf, pages 8.

For Octave references are used two main sources : Octave.org and Octave.sourceforge.io

https://octave.sourceforge.io/octave/function/disp.html

https://octave.sourceforge.io/octave/function/close.html

https://octave.sourceforge.io/octave/function/clear.html

https://octave.sourceforge.io/octave/function/find.html

https://octave.sourceforge.io/octave/function/mean.html

https://octave.sourceforge.io/octave/function/log.html

https://docs.octave.org/v8.4.0/Manipulation-of-Plot-Windows.html

https://docs.octave.org/v8.4.0/Manipulation-of-Plot-Windows.html

https://octave.sourceforge.io/octave/function/subplot.html

https://docs.octave.org/v8.4.0/Plot-Annotations.html#XREFylabel

https://octave.sourceforge.io/octave/function/xlim.html

https://octave.sourceforge.io/octave/function/scatter.html

https://octave.sourceforge.io/octave/function/set.html

https://docs.octave.org/v8.4.0/Using-Packages.html

https://octave.sourceforge.io/octave/function/length.html

https://octave.sourceforge.io/octave/function/transpose.html

https://octave.sourceforge.io/octave/function/corr.html

https://octave.sourceforge.io/octave/function/print.html

https://octave.sourceforge.io/octave/function/addpath.html

https://www.dynare.org/resources/quick_start/#running-and-editing-a-dynare-example-on-windows

https://docs.octave.org/latest/Multiple-Plot-Windows.html

https://docs.octave.org/v6.2.0/Arithmetic-Ops.html

https://docs.octave.org/v6.2.0/Arithmetic-Ops.html

https://docs.octave.org/latest/Printing-and-Saving-Plots.html

Advanced Macroeconomics Lecture : Chapter 7 , (hpfilter.m)

Dynare Resources :

https://www.dynare.org/manual/the-model-file.html
Section4.2
https://www.dynare.org/manual/the-model-file.html
Section 4.2
https://www.dynare.org/manual/the-model-file.html
Section4.2
https://www.dynare.org/manual/the-model-file.html
Section4.2.1
https://www.dynare.org/manual/the-model-file.html
Section 4.7

https://www.dynare.org/manual/the-model-file.html
Section 4.8
https://www.dynare.org/manual/the-model-file.html
Section 4.10.1
https://www.dynare.org/manual/the-model-file.html
Section 4.13.1

Declaration of independence/assurance:

“With my signature I declare that I have worked on the home assignment independently and without the
use of unauthorised aids or the help of other persons.”

22.02.2024 Erida Gjana (222221447)